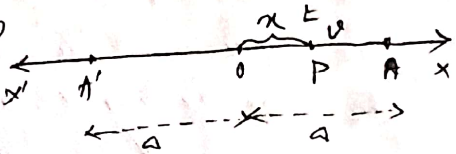
Simple Harmonic Motion

1. When a particle moves in a st. line so that its acceleration at any point of its motion is always directed to a fixed point on the line and is proportional to its distance from a fixed point, prove that the particle is said to execute simple harmonic motion. Also mention the frequency and amplitude of the simple harmonic motion. 22

Ans. Let $x'Ox$ be the st. line along which a particle executes S.H.M with O as its mean position towards which the acceleration is always directed.



Let P be the position of the particle at time t. The acceleration of the particle is μx directed towards O, μ being a constant.

Equation of motion of the particle of mass m is

$$m\ddot{x} = -m\mu x$$

$$\text{or } \ddot{x} = -\mu x \quad \text{--- (1)}$$

Since acceleration is in the opposite direction in which x increases, equation (1) is true for either $x>0$ or $x<0$ on either side of O. Multiplying both sides of eqn (1) by $2\dot{x}$ and integrating, we have

$$\dot{x}^2 = -\mu x^2 + C \quad \text{--- (2)}$$

If at $t=0$, the particle is at $x=a$, i.e. initially motion starts from rest at A, where $OA = a$, we have

$$0 = -\mu a^2 + C$$

$$\Rightarrow C = +\mu a^2$$

$$\therefore \dot{x}^2 = -\mu x^2 + \mu a^2 = \mu(a^2 - x^2) \quad \text{--- (3)}$$

$$\text{or, } \dot{x} = v = \pm\sqrt{\mu}\cdot\sqrt{a^2 - x^2}$$

Now the velocity is clearly negative as long as OP is positive. So, we take only (-ve) sign.

$$\dot{x} = -\sqrt{\mu}\sqrt{a^2 - x^2} \quad \text{--- (4)}$$

Integrating ④ gives

$$-\int \frac{dx}{\sqrt{a^2-x^2}} = \sqrt{\mu}\int dt$$

or, $\sqrt{\mu}\,t = \cos^{-1} x/a + c_1$

Now $x = a$ at $t = 0$ at the starting point A.

$\therefore 0 = \cos^{-1} 1 + c_1 \quad \therefore$ i.e. $c_1 = 0$

Hence $x = a\cos\sqrt{\mu}\,t$ - - ⑤

Equation ④ & ⑤ give the velocity & displacement for motion from A to O. When the particle reaches O, the velocity $= -a\sqrt{\mu}$ and the time from A to O is given by putting $x = 0$ in ⑤, which gives $t = \frac{\pi}{2\sqrt{\mu}}$.

Evidently, as soon as the particle crosses O and moves along OX' the acceleration alters its direction and tends to diminish the velocity which is destroyed as rapidly as it was produced during its motion from A to O. The particle comes to rest at A' such that $OA' = OA = a$, the particle then reaches the path from A' to O and comes to instantaneous rest at A. The oscillatory motion from A to A' and back is continued repeatedly over and over again. It is clear from ⑤ that the particle attains the same displacement after intervals of $2\pi/\sqrt{\mu}$, $4\pi/\sqrt{\mu}$, $6\pi/\sqrt{\mu}$ ...

Since ⑤ can be written as

$$x = a\cos\sqrt{\mu}\left(t + \frac{2\pi K}{\sqrt{\mu}}\right)$$

K being an integer.

The time $2\pi/\sqrt{\mu}$ is called the period of oscillation of the S.H.M, evidently it is four times the time from A to O. The frequency of oscillation, n, defined as the number of complete oscillation per unit time, is given by $n = \sqrt{\mu}/2\pi$.

The distance 'a' (OA or OA') of extreme points from the centre of oscillation is called the amplitude of the S.H.M.

Q. a) A particle moves along a st. line under the law of motion given by $x = a\cos(nt+b)$. Show that the acceleration is directed to the origin and varies as the distance.

b) A particle is performing a simple harmonic motion of period T about a centre O and it passes through a point P with a velocity v in the direction OP. If the particle returns to P in time t, then show that
$$t = \frac{T}{\pi}\tan^{-1}\left(\frac{vT}{2\pi x}\right) \text{ or } OP = \frac{vT}{2\pi}\cot\frac{\pi t}{T}, \text{ where } OP = x.$$
[20]

Ans a) Here, the displacement x of the particle at any time t is given by,
$$x = a\cos(nt+b) \quad \text{---} ①$$
d.w.r. to t
$$\dot{x} = -an\sin(nt+b)$$
d.w.r. to t
$$\ddot{x} = -an^2\cos(nt+b)$$
$$\text{or } \ddot{x} = -n^2 x \quad \text{by } ①$$

So, the acceleration varies as the distance x of the particle from the origin and the negative sign indicates that the acceleration is directed towards the origin.

b) The equation of motion of a particle executing S.H.M is
$$\ddot{x} = -\mu x$$
with time period $T = \frac{2\pi}{\sqrt{\mu}}$.

and the displacement x is given by
$$x = a\sin\sqrt{\mu}t$$
where a is the amplitude and the particle was at O at $t=0$.

Let the particle be at P at time t_1 and come to P again at time $t+t_1$.

Then $x = a\sin\sqrt{\mu}t_1$ --- ①

and $x = a\sin\sqrt{\mu}(t+t_1)$ --- ②

from ① $\frac{dx}{dt_1} = a\sqrt{\mu}\cos\sqrt{\mu}t_1 = v$ given --- ③

from ④ $\Rightarrow$ $x = a\sin\sqrt{\mu}t \cdot \cos\sqrt{\mu}t_1 + a\cos\sqrt{\mu}t \cdot \sin\sqrt{\mu}t_1$

$$= \frac{v}{\sqrt{\mu}}\sin\sqrt{\mu}t + x\cos\sqrt{\mu}t \quad [\text{by } ① \,\&\, ③]$$

or, $x(1-\cos\sqrt{\mu}t) = \frac{v}{\sqrt{\mu}}\sin\sqrt{\mu}t$

or, $x\cdot\cancel{2}\sin^2\frac{\sqrt{\mu}t}{2} = \frac{v}{\sqrt{\mu}}\cancel{2}\sin\frac{\sqrt{\mu}t}{2}\cdot\cos\frac{\sqrt{\mu}t}{2}$

or $\tan\frac{\sqrt{\mu}t}{2} = \frac{v}{x\sqrt{\mu}}$

or, $\frac{1}{2}\sqrt{\mu}t = \tan^{-1}\frac{v}{x\sqrt{\mu}}$

i.e. $t = \frac{2}{\sqrt{\mu}}\tan^{-1}\frac{v}{x\sqrt{\mu}}$

So, $t = \frac{T}{\pi}\tan^{-1}\frac{vT}{2\pi x}$, Since $T = \frac{2\pi}{\sqrt{\mu}}$

Since $\frac{t\pi}{T} = \tan^{-1}\frac{vT}{2\pi x}$

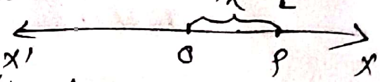
$$\tan\frac{t\pi}{T} = \frac{vT}{2\pi x}$$

or, $\cot\frac{t\pi}{T} = \frac{2\pi x}{vT}$

Hence, $x = \frac{vT}{2\pi}\cot\frac{\pi t}{T}$.

3.8 A particle is moving in a st. line with an acceleration n^2x towards a fixed ~~point~~ origin on the line and is simultaneously acted on by a periodic force per unit mass. Discuss the motion when $n \neq p$.

Ans. Let P be the position of the particle, at any time t, moving along the st. line OX, under a force mn^2x towards the fixed point O and that $OP = x$. At the same time, it is also acted upon by a periodic force $mf\cos pt$. Then the equation of motion is



$$m\ddot{x} + mn^2x = mf\cos pt$$

$$\ddot{x} + n^2x = f\cos pt \quad --- (1)$$

The general solution of this equation consists of a complementary function and a particular integral. The complementary function is

$$A\cos nt + B\sin nt = a\cos(nt+\epsilon)$$

The particular integral is

$$\frac{1}{D^2+n^2} F \cos pt$$

$$= \frac{1}{-p^2+n^2} F \cos pt \qquad \text{where } p \neq n$$

$$= \frac{1}{n^2-p^2} F \cos pt$$

Thus the general solution is of the form

$$x = a \cos(nt+\epsilon) + \frac{F}{n^2-p^2} \cos pt \quad \text{---} (2)$$

Where a and ϵ are constants to be determined from the initial conditions.

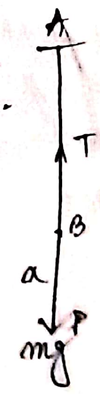
Thus, the motion of the particle is the resultant of two simple harmonic motion of periods $2\pi/n$ and $2\pi/p$. The first part is the usual S.H.M in the absence of the periodic disturbing force and is known as free oscillation and the second part is simply a superposed S.H.M. and is known as forced oscillation.

When n^2-p^2 is very small, then the forced oscillation represented by

$$\frac{F}{n^2-p^2} \cos pt$$

will be very large and the particle will oscillate with a period which is practically the same as that of the forced oscillation.

4. One end of an elastic string of length a is fixed at A and the other end is fastened to a heavy particle, the modulus of elasticity of the string being equal to the weight of the particle. Show that, if the particle be dropped from A, then it will descend a distance $(2+\sqrt{3})a$ before coming to rest. [20]



An Let $AB = a$ and also let P be the position of the particle after time t, where $AP = x$.

If T be the tension of the string in this position.

$$T = \lambda \cdot \frac{x-a}{a} = mg \cdot \frac{x-a}{a}, \text{ Since } \lambda = \text{wt. of the particle} = mg$$

Equation of motion of the particle is

$$mv\frac{dv}{dx} = mg - T = mg - mg\frac{x-a}{a}$$

or, $$v\frac{dv}{dx} = -\frac{g}{a}(x-2a) \quad --- (1)$$

Thus the motion of the particle is simple harmonic about $x-2a=0$ i.e. $x=2a$, the equilibrium position.

From (1), we have on integration

$$v^2 = -\frac{g}{a}(x-2a)^2 + A \quad —— (2)$$

where A is the constant of integration.

Obviously, the S.H.M starts when the particle B i.e. when it has fallen a distance a freely under gravity; and the velocity of the mass acquired due to fall is $\sqrt{2ga}$.

So, initially, at $t=0$, $v=\sqrt{2ga}$, when $x=AB=a$

from (2) $$2ga = -\frac{g}{a}(a-2a)^2 + A$$

$$\Rightarrow A = 3ag.$$

Thus (2) $\Rightarrow$ $$v^2 = -\frac{g}{a}(x-2a)^2 + 3ag$$

The particle comes to rest, when $v=0$

i.e. $$0 = -\frac{g}{a}(x-2a)^2 + 3ag$$

or, $$(x-2a)^2 = 3a^2$$

or, $$x-2a = \pm\sqrt{3}a$$

Rejecting the negative sign,

$$\boxed{x = (2+\sqrt{3})a}$$

§ The speed v of a point moving along the x-axis is given by $v^2 = 16-x^2$. Prove that the motion is simple harmonic. Find also amplitude.

Ans. Here $v^2 = 16-x^2$ —— (1)

d.w.r. to x, we have

$$2v\frac{dv}{dx} = -2x$$

i.e. $$v\frac{dv}{dx} = -x$$

or, $\ddot{x} = -x$ —— (2)

Equation of motion (2) indicates that the motion of the particle is simple harmonic motion.

Again, from (1) $v=0$, when $16-x^2=0$

i.e. $x=\pm 4$

So, amplitude of motion is $\dfrac{4-(-4)}{2} = 4$.

§ If the velocity of a particle moving along the x-axis is given by $v^2 = 8bx - x^2 - 12b^2$. Show that the motion is simple harmonic.

Ans. Since $v^2 = 8bx - x^2 - 12b^2$

D.w.r. to x

$$2v\frac{dv}{dx} = 8b - 2x$$

or, $\ddot{x} = 4b - x = -(x-4b)$ —— (1)

If we put $z = x-4b$, $\dfrac{dz}{dt} = \dfrac{dx}{dt}$, $\dfrac{d^2z}{dt^2} = \dfrac{d^2x}{dt^2}$

So, equ (1) becomes $\dfrac{d^2z}{dt^2} = -z$

This indicates that the motion of the particle is simple harmonic motion about the mean position $z=0$ i.e. $x=4b$.

§ A particle moves along a st. line under the law of force given by $x = a\cos(nt+b)$. Show that the acceleration is directed towards the origin and varies as the distance.

Ans. Here, the displacement x of the particle at any time t is given by

$$x = a\cos(nt+b) \quad \text{—— (1)}$$

So, $\dot{x} = -an\sin(nt+b)$

and $\ddot{x} = -an^2\cos(nt+b)$

$\ddot{x} = -n^2x$ [by (1)]

So, the acceleration varies as the distance x of the particle from the origin and the negative sign indicates that the acceleration is directed towards the origin.

§ The distance of a particle, performing simple harmonic motion, from the middle point of its path at three consecutive seconds are observed to be x, y, z. Show that the time of a complete oscillation is $\dfrac{2\pi}{\cos^{-1}\left(\frac{x+z}{2y}\right)}$

Ans. Let the distances of the particle from the middle point of its path at $(t-1)$, t and $(t+1)$ be x, y and z resp.

If the equation of motion of the particle is

$$\ddot{x} = -\mu x \quad (\mu > 0)$$

and a be the amplitude of motion.

$$\therefore \quad x = a\cos\sqrt{\mu}(t-1) \quad \text{--- } (1)$$

$$y = a\cos\sqrt{\mu}t \quad \text{--- } (2)$$

$$z = a\cos\sqrt{\mu}(t+1) \quad \text{--- } (3)$$

Now, $x+z = a\cos\sqrt{\mu}(t-1) + a\cos\sqrt{\mu}(t+1)$

$$= 2a\cos\sqrt{\mu}t \cdot \cos\sqrt{\mu}$$

$$= 2 \cdot y \cdot \cos\sqrt{\mu}$$

or $\cos\sqrt{\mu} = \dfrac{x+z}{2y}$

i.e. $\sqrt{\mu} = \cos^{-1}\left(\dfrac{x+z}{2y}\right)$

Time of a complete oscillation

= periodic time

$$= T = \frac{2\pi}{\sqrt{\mu}} = \frac{2\pi}{\cos^{-1}\frac{x+z}{2y}}$$